Muhammad Uns Haider Shah

Scientific Machine Learning · Computational Fluid Dynamics · Aerodynamic Design Optimization · Diagnostics & Prognostics
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RESEARCH INTERESTS

Scientific machine learning for engineering systems: surrogate and reduced-order modeling, Bayesian optimization, and neural-operator and physics-informed methods for aerodynamic design and flow-field prediction; the trustworthiness of predicted uncertainty in data-driven models; and machine-learning diagnostics and prognostics for rotating machinery and propulsion systems. Seeking a fully funded, thesis-based graduate position (MS / MASc) in mechanical or aerospace engineering beginning Fall 2027.

PUBLICATIONS

Design and Optimization of a Blended-Wing-Body UAV using AI-Driven Surrogate Modeling and CFD Analysis
M. S. Naseem, **M. U. H. Shah**, et al. *Proceedings of the IMechE, Part G: Journal of Aerospace Engineering* (SCI / Scopus-indexed), 2026. DOI: 10.1177/09544100261447561

- My contribution: developed the machine-learning surrogate and optimization pipeline end to end. Built a Gaussian Process Regression surrogate with Expected-Improvement Bayesian optimization over DOE-structured RANS CFD data to optimize wing planform, airfoil, winglet and twist; the surrogate's uncertainty estimates drove adaptive sampling. Cruise lift-to-drag ratio rose from 9.9 to 12.2 (about 23%) in about 80 CFD evaluations.
- Conducted the 3D Reynolds-averaged Navier–Stokes CFD (ANSYS Fluent, k-ω SST) that generated the aerodynamic training data and verified the optimized configuration.
- The wider team project also spanned CAD, finite-element analysis, carbon-fibre composite prototyping and wind-tunnel testing.

CONFERENCE PRESENTATIONS

Oral Presentation, IBCAST 2025 Aug 2025
22nd International Bhurban Conference on Applied Sciences & Technology, Fluid Dynamics track. Presented the blended-wing-body UAV optimization work; camera-ready paper submitted to the proceedings (IEEE Xplore), awaiting publication.

EDUCATION

B.Sc. in Mechanical Engineering Sep 2021 – Jul 2025
COMSATS University Islamabad (Wah Campus), Pakistan HEC- and PEC-recognized

- Relevant coursework: Computational Fluid Dynamics (A–), Heat & Mass Transfer (A–), CAD/CAM (A–), Instrumentation & Measurement (A–), Maintenance Engineering (A), Mechanisms & Mechanical Vibrations Lab (A), Numerical Computations (B+), Fluid Mechanics I & II, Finite Element Analysis, Mechanical Vibrations, Control Engineering, Machine Design, IC Engines.

RESEARCH PROJECTS: AERODYNAMIC DESIGN & TRUSTWORTHY SURROGATES

When Can a Data-Driven Surrogate Be Trusted? Airfoil Shape Optimization on AirfRANS Aug – Sep 2026
Python, PyTorch, scikit-learn, Gaussian processes · Code

- Trained neural surrogates on 2D steady RANS solutions over NACA airfoils (Re 2–6 M): a global shape descriptor cut surface-pressure error 15-fold and raised lift rank correlation to 0.998; showed field-integrated drag is unrankable (rank correlation 0.075) because the ~2 μm near-wall layer is unresolved.
- Direct force surrogates on 200 held-out simulations: the Gaussian process reached 0.3% median drag error with 91% of log-drag predictions inside its ±2σ band, against 55% for a neural ensemble.
- Drove shape optimization with the surrogate: Bayesian and automatic-differentiation gradient search both reached within 0.33% of a 200,000-point reference optimum, but retraining moved the optimum by 14–20% of the design range, and predicted uncertainty under-estimated error 139× at a likely separated-flow case.

RESEARCH PROJECTS: MACHINE LEARNING FOR MACHINERY HEALTH

Turbofan Remaining-Useful-Life Prediction · NASA C-MAPSS Mar – May 2026
Python, scikit-learn · Code

- Random-Forest RUL model on FD001 reached RMSE 17.97 / MAE 12.74 cycles (14.1% RMSE and 22.3% MAE improvement over a linear baseline), in line with published classical and early deep-learning baselines.
- Leakage-safe, per-engine 5-cycle rolling features turned sensor 4 from 0.01 to 0.63 feature importance; K-Means (k = 3) on PCA-reduced profiles separated distinct degradation pathways.

Rolling-Bearing Fault Diagnosis · CWRU Vibration Dataset Oct – Dec 2025

Python, scikit-learn · Code

- Classified 10 bearing conditions from 48 kHz drive-end vibration (9 time-domain features over 2048-point windows): 96.4% ± 0.8% under stratified 5-fold cross-validation and 92.2% on a held-out split.
- Traced the remaining errors to fault classes whose kurtosis stays near the healthy level, and documented the optimism of window-level splits and the recording-offset shortcut in the mean feature.

PROFESSIONAL EXPERIENCE

Independent Researcher Jul 2025 – Present

Islamabad, Pakistan

- Self-directed research on the reliability of machine-learning surrogates for engineering design: uncertainty calibration, extrapolation, and the stability of surrogate-predicted optima.
- Machine learning for CFD: neural and Gaussian-process surrogates trained on RANS simulations (AirfRANS benchmark) to drive airfoil shape optimization; code and technical write-up public on GitHub.

CMMS Engineer, Asset Reliability & Predictive Maintenance Jan – Aug 2026

Invotyx (ADNOC Programme), Islamabad

- Built and validated ISO 14224-compliant asset hierarchies and SAP PM master data for rotating and static equipment, supporting reliability-centred and predictive-maintenance workflows.
- Extracted technical attributes from P&IDs and vendor datasheets; enforced CMMS data-governance rules across live multi-discipline asset registers.

Aerospace B1 Engineering Intern, Line Maintenance Jul – Aug 2024

Pakistan International Airlines (PIA)

- Performed transit, daily and cabin checks across ATR-42/72, A320 and B-777 fleets; studied AMM/MEL/TechLog systems and glass-cockpit avionics.

SPECIALIZED TRAINING

Machine Learning and AI for Fluid Dynamics, FlowThermoLab Jun 2026 – present

Certification cohort led by Prof. Ricardo Vinuesa: turbulence prediction, reduced-order modeling (POD/DMD, autoencoders), flow control and optimization, transformers for flow fields, physics-informed neural networks and neural operators (FNO, DeepONet), in Python and PyTorch.

Professional Data Analytics Certification, Institute of Emerging Careers Nov 2025 – May 2026

Applied data analysis with Excel, SQL, Power BI and Python.

ENGINEERING & FABRICATION PROJECTS

- **CFRP composite manufacturing:** designed 3D-printed split moulds (SolidWorks) and produced the BWB-UAV airframe skins in carbon-fibre/epoxy by vacuum-bag resin infusion; the prototype was then force-and-moment tested in a wind tunnel.
- **Shell-and-tube heat exchanger:** designed (LMTD / Dittus–Boelter / Bell–Delaware), fabricated and tested a 7 ft baffled unit; the measured energy balance agreed with the prediction to within 5%.

TECHNICAL SKILLS

Scientific ML & surrogate modeling: Gaussian Process Regression, Bayesian optimization, Design of Experiments, uncertainty quantification and calibration, neural ensembles, automatic differentiation for design sensitivities; PINNs and neural operators (in development).
Machine learning & programming: Python (PyTorch, scikit-learn, pandas, NumPy, Matplotlib/Seaborn); Random Forests, K-Means/PCA, feature engineering, cross-validation.
Computational & CAE: ANSYS Fluent (CFD), ANSYS Mechanical (FEA), SolidWorks, AutoCAD.
Signals, prognostics & reliability: vibration health indicators (RMS, kurtosis, crest factor), degradation feature engineering, RUL regression; ISO 14224, SAP PM master data.
Tools: Git/GitHub, LaTeX, SQL (MySQL), Power BI (DAX), Arduino (C++).

LEADERSHIP, AWARDS & AFFILIATIONS

Planning Head & Treasurer, ASME COMSATS Wah Student Section Jul 2023 – Sep 2025

- Part of the leadership team of a 100+ member section: ran workshops, seminars and inter-university competitions; owned the budget and financial reporting. Voted Best Member of the Year, 2023–24.
- Affiliations: Member (Early Career), ASME (2026–present); student member 2023–2025 of AIAA, RAeS, IMechE, ASME and ASEE. Registered Engineer, Pakistan Engineering Council.